

**SIMATS ENGINEERING
ASSESSMENT TOOL 1**

COURSE NAME: Artificial Intelligence COURSE CODE: CSA17

COURSE OUTCOME COVERED

***CO3:** Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4
Marks:40

Total

Annexure A

CASE STUDY

Code of Conduct:

I A Sai Swetha (192319072) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

- I. If a patient has Fever AND Cough $\rightarrow$ Possible Flu
- ii. If a patient has Fever AND Rash $\rightarrow$ Possible Measles
- iii. If a patient has Cough AND Breathlessness $\rightarrow$ Possible Pneumonia
- iv. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

- (a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.
- (b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.
- (c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

(a) Propositional Logic representation

Let F = Patient A has Fever, C = Patient A has Cough, B = Patient A has Breathlessness, Fl = Patient A has Flu, M = Patient A has Measles, and P = Patient A has Pneumonia.

Rule 1: $F \wedge C \rightarrow Fl$

Rule 2: $F \wedge \text{Rash} \rightarrow M$, where Rash denotes the patient having a rash.

Rule 3: $C \wedge B \rightarrow P$

Facts: F = True, C = True, B = True.

The arrow $\rightarrow$ denotes implication, $\wedge$ denotes logical AND, and $\neg$ denotes logical NOT.

(b) Forward Chaining

Initial facts: Fever, Cough, Breathlessness are true.

Rule 1 fires: Fever $\wedge$ Cough is true $\rightarrow$ derive Flu.

Rule 2: Fever is true, but Rash is not given $\rightarrow$ Measles cannot be derived.

Rule 3 fires: Cough $\wedge$ Breathlessness is true $\rightarrow$ derive Pneumonia.

Final conclusions: Patient A has Flu and Pneumonia. Measles is not derivable from the supplied facts.

(c) Backward Chaining for 'Patient A has Pneumonia'

Goal: Pneumonia(A)

Goal reduction: $\text{Pneumonia}(A) \leftarrow \text{Cough}(A) \wedge \text{Breathlessness}(A)$

Cough(A) is a known fact. Breathlessness(A) is a known fact. Therefore both subgoals succeed, so Pneumonia(A) is proved.

Goal tree:

Pneumonia(A)

├— Cough(A) ✓ fact

└— Breathlessness(A) ✓ fact

Therefore: Pneumonia(A) ✓

Case Study 2 — Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

- I. $\forall x: \text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$ [Rule 1]
- II. $\forall x: \text{ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$ [Rule 2]
- III. $\forall x \forall y: \text{Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$ [Rule 3]
- IV. Facts: $\text{Vehicle}(\text{Ambulance}), \text{EmergencyType}(\text{Ambulance}), \text{Vehicle}(\text{CarA}), \text{Behind}(\text{CarA}, \text{Ambulance})$

(a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.

(b) Using the Resolution method, prove that 'Proceed (CarA)' is true. Convert relevant clauses to CNF and show each resolution step.

(c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

(a) Unification

Rule 1: $\forall x [\text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)]$

For the fact $\text{EmergencyType}(\text{Ambulance})$, substitute $x = \text{Ambulance}$. The same substitution matches $\text{Vehicle}(\text{Ambulance})$.

MGU (Most General Unifier): $\theta = \{x/\text{Ambulance}\}$

After substitution: $\text{Vehicle}(\text{Ambulance}) \wedge \text{EmergencyType}(\text{Ambulance}) \rightarrow \text{ClearPath}(\text{Ambulance})$.

Both antecedents are facts, so $\text{ClearPath}(\text{Ambulance})$ follows.

(b) Resolution proof of Proceed(CarA)

The supplied KB supports $\text{ClearPath}(\text{Ambulance})$, then $\text{GreenSignal}(\text{Ambulance})$, and hence $\text{Proceed}(\text{CarA})$ using Rule 3.

Relevant clauses in CNF:

C1: $\neg \text{Vehicle}(x) \vee \neg \text{EmergencyType}(x) \vee \text{ClearPath}(x)$

C2: $\neg \text{ClearPath}(x) \vee \text{GreenSignal}(x)$

C3: $\neg \text{ClearPath}(x) \vee \neg \text{GreenSignal}(x) \vee \neg \text{RedSignal}(x)$

C4: $\neg \text{Vehicle}(x) \vee \neg \text{Behind}(x,y) \vee \neg \text{GreenSignal}(y) \vee \text{Proceed}(x)$

C5: $\text{Vehicle}(\text{Ambulance})$

C6: $\text{EmergencyType}(\text{Ambulance})$

C7: $\text{Vehicle}(\text{CarA})$

C8: $\text{Behind}(\text{CarA}, \text{Ambulance})$

Resolution sequence:

C1 with C5 and C6, using $x = \text{Ambulance} \rightarrow \text{ClearPath}(\text{Ambulance})$.

C2 with $\text{ClearPath}(\text{Ambulance}) \rightarrow \text{GreenSignal}(\text{Ambulance})$.

C4 with $\text{Vehicle}(\text{CarA})$, $\text{Behind}(\text{CarA}, \text{Ambulance})$, and $\text{GreenSignal}(\text{Ambulance}) \rightarrow \text{Proceed}(\text{CarA})$.

Thus $\text{Proceed}(\text{CarA})$ is entailed by the given rules and facts.

(c) Two simultaneous emergency vehicles

The current KB is sufficient to reason about each emergency vehicle independently when each has $\text{Vehicle}(x)$ and $\text{EmergencyType}(x)$, because Rule 1 is universally quantified. However, it does not explicitly model priority or conflict when two emergency vehicles require the same path or signal.

Additional knowledge that would improve the model:

A rule assigning priority between emergency vehicles, e.g. based on emergency severity.

Rules for two emergency vehicles requesting the same route or intersection.

Facts identifying the emergency type/severity and current location of each emergency vehicle.

Rules ensuring mutually conflicting signals are not simultaneously assigned to the same traffic movement.

Case Study 3 — Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

- 1) $P1: \text{SoilDry} \rightarrow \text{IrrigationNeeded}$
- 2) $P2: \text{IrrigationNeeded} \wedge \text{CropWheat} \rightarrow \text{ApplyDripMethod}$
- 3) $P3: \neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$
- 4) Facts: $\text{SoilDry} = \text{True}$, $\text{CropWheat} = \text{True}$

(a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication $\rightarrow$ move $\neg$ inward $\rightarrow$ distribute).

(b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause $\square$. Draw the resolution proof tree.

(c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

(a) CNF conversion

Let $S = \text{SoilDry}$, $I = \text{IrrigationNeeded}$, $W = \text{CropWheat}$, $D = \text{ApplyDripMethod}$, $A = \text{CropAtRisk}$.

$P1: S \rightarrow I$

Eliminate implication: $\neg S \vee I$. This is already CNF.

$P2: I \wedge W \rightarrow D$

Eliminate implication: $\neg(I \wedge W) \vee D$. Move negation inward: $(\neg I \vee \neg W) \vee D$. CNF clause: $\neg I \vee \neg W \vee D$.

$P3: \neg D \wedge W \rightarrow A$

Eliminate implication: $\neg(\neg D \wedge W) \vee A$. Move negation inward: $(D \vee \neg W) \vee A$. CNF clause: $D \vee \neg W \vee A$.

Facts in CNF: S and W.

(b) Resolution Refutation for ApplyDripMethod

Goal: D. Negate the goal: $\neg D$.

C1: $\neg S \vee I$

C2: $\neg I \vee \neg W \vee D$

C3: $D \vee \neg W \vee A$

C4: S

C5: W

C6: $\neg D$ (negated goal)

Resolution steps:

Resolve C1 with C4 on S $\rightarrow$ I.

Resolve I with C2 on I $\rightarrow \neg W \vee D$.

Resolve $\neg W \vee D$ with C5 on W $\rightarrow D$.

Resolve D with C6 on D $\rightarrow \square$ (empty clause).

Since the empty clause is derived, the negation of the goal is inconsistent with the KB. Therefore ApplyDripMethod is proved.

(c) Removing SoilDry

If S is removed, C4 is no longer available. Therefore P1 cannot fire, so IrrigationNeeded (I) cannot be derived from the remaining facts and rules. Without I, P2 cannot derive ApplyDripMethod (D).

CropAtRisk (A) also does not follow. P3 requires $\neg$ ApplyDripMethod and CropWheat. CropWheat remains true, but $\neg$ ApplyDripMethod is not a supplied fact and cannot be inferred merely because D is no longer derivable. Therefore A is not logically entailed from the remaining KB.

Case Study 4 — Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

- 1) Stench [1,2] $\rightarrow$ Wumpus is adjacent to cell [1,2]
- 2) Breeze [1,1] $\rightarrow$ Pit is adjacent to cell [1,1]
- 3) Glitter [x,y] $\rightarrow$ Gold is at [x,y]
- 4) $\neg$ Wumpus [x,y] $\wedge$ $\neg$ Pit[x,y] $\rightarrow$ Safe[x,y]

(a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.

(b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.

(c) Evaluate whether the path [1,1] $\rightarrow$ [1,2] $\rightarrow$ [2,2] is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

(a) Belief state and Modus Ponens

Percepts: Stench[1,2], Breeze[1,1], Glitter[2,2].

Using the supplied rules:

Stench[1,2] $\rightarrow$ Wumpus is adjacent to [1,2]. Therefore the belief state contains Wumpus-adjacent([1,2]).

Breeze[1,1] $\rightarrow$ Pit is adjacent to [1,1]. Therefore the belief state contains Pit-adjacent([1,1]).

Glitter[2,2] $\rightarrow$ Gold[2,2]. Therefore Gold is at [2,2].

The rule $\neg\text{Wumpus}[x,y] \wedge \neg\text{Pit}[x,y] \rightarrow \text{Safe}[x,y]$ cannot be fired for a cell unless both the corresponding negative facts are known. The supplied percepts alone do not provide those negative facts.

(b) Forward Chaining – possible Wumpus and Pit locations

From Stench[1,2], the Wumpus must be in a cell adjacent to [1,2]. From Breeze[1,1], a pit must be in a cell adjacent to [1,1]. The exact neighboring cells cannot be uniquely identified from the simplified rules as written.

Thus the precise conclusions supported by the supplied KB are:

Wumpus is adjacent to [1,2].

Pit is adjacent to [1,1].

Gold is at [2,2].

No exact Wumpus/Pit cell is uniquely derivable without additional adjacency and exclusion rules.

(c) Safety of path [1,1] $\rightarrow$ [1,2] $\rightarrow$ [2,2]

The path cannot be certified as completely safe from the supplied KB. In particular, Stench[1,2] indicates that a Wumpus is adjacent to [1,2], and Breeze[1,1] indicates that a pit is adjacent to [1,1]. The KB does not establish $\neg$ Wumpus and $\neg$ Pit for the cells on the path.

Therefore the logically justified conclusion is: the path is not proven safe. The agent should not classify it as safe using only the supplied knowledge.

RUBRICS FOR CALCULATION

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand